MILITARY SPECIFICATION SHEET
FILTERS, RADIO FREQUENCY INTERFERENCE
STYLE FL42

The Part or Identifying Numbers (PIN) M15733/28-0003 and -0004 were made inactive for new design as of 13 August 1979. PIN's M15733/28-0001 and -0002 were made inactive for new design as of 29 November 1982. For new design, see table II.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-15733.

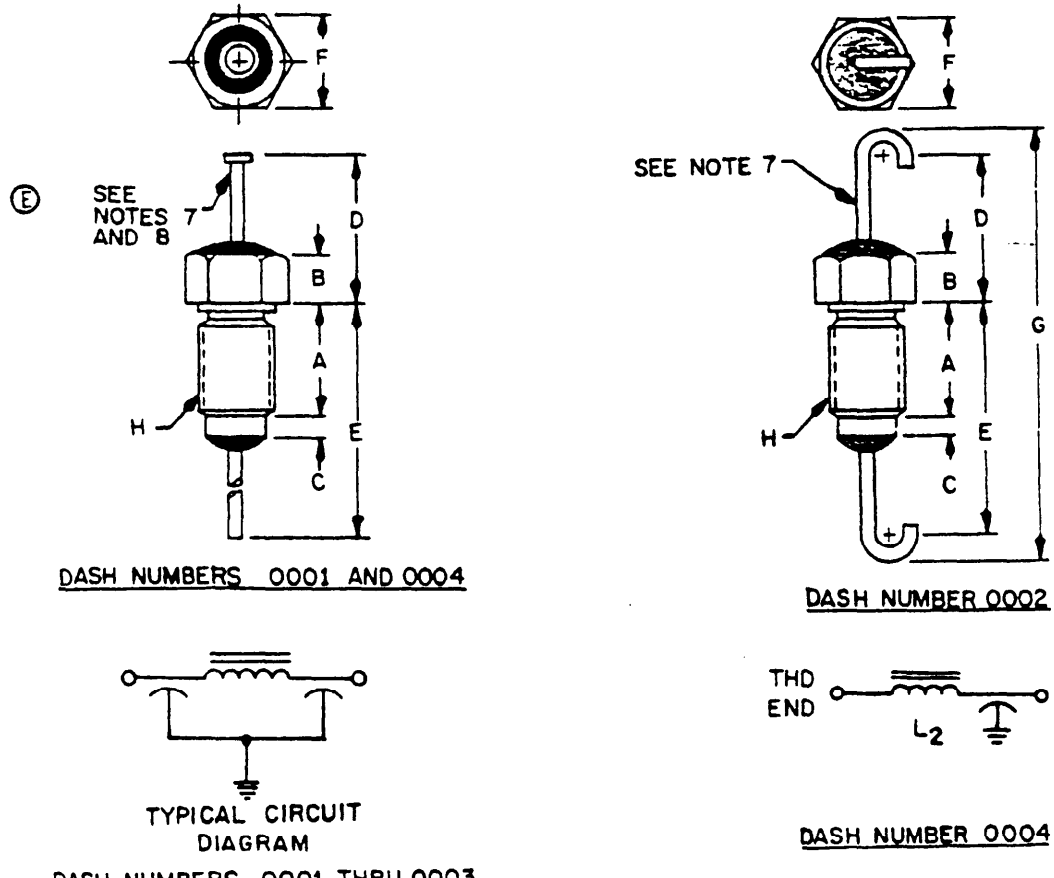
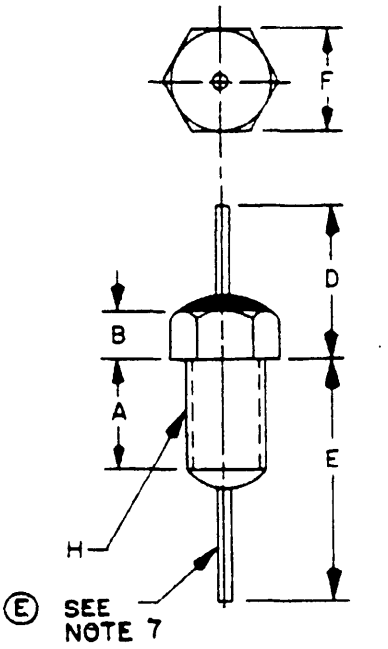


FIGURE 1. Case and hardware dimensions and circuit configuration.

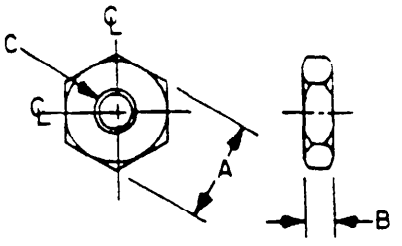
(E) denotes change

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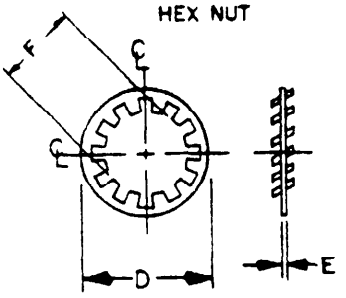
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DASH NUMBER 0003



HEX NUT



LOCKWASHER

Inches	mm
.025	0.64
.032	0.81
.040	1.02
.060	1.52

MOUNTING HARDWARE

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. Mounting torque 3 inch-pounds maximum.
5. Case is ground terminal.
6. Mounting hardware (lockwasher and hex nut) will be supplied with filter.
7. Leads shall be solid, tinned, AWG 20 (.032 dia) for dash numbers 0001 and 0002, and AWG 22 (.025 dia) for dash number 0003.
8. Leads shall be solid, copper, silver plated, AWG 18 (.040 dia) for dash number 0004.
9. Terminal identification (non-symmetrical filters): The case shall be marked at the threaded end of the filter, with the symbol "C" or "L", as follows:

Circuit	Symbol
L_1	C
L_2	L

10. Turret headed terminal shown for dash numbers 0001 and 0004 is optional.
Ⓔ 11. For dash number 0003, an undercut or imperfect threads out to a maximum of .060 inch (1.52 mm) from the hex head or mounting surface is permissible.

FIGURE 1. Case and hardware dimensions and circuit configuration - Continued.

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Filter dimensions and weights.

Type designation 1/ 2/	Dash no.	A		B		C		D		E		F		G		H	Weight (grams)
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Thread	
FL42GW1ED3	0001	.201 (5.11)	.231 (5.87)	.078 (1.98)	.108 (2.74)	.035 (.89)	.065 (1.65)	.281 (7.14)	.343 (8.71)	.859 (21.82)	.921 (23.39)	.172 (4.37)	.202 (5.13)	---	---	.164-32 UNC-2A	1.7
FL42GW1ED3	0002	.201 (5.11)	.231 (5.87)	.078 (1.98)	.108 (2.74)	.035 (.89)	.065 (1.65)	.281 (7.14)	.343 (8.71)	.435 (11.05)	.515 (13.08)	.172 (4.37)	.202 (5.13)	.875 (22.23)	.937 (23.80)	.164-32 UNC-2A	1.7
FL42GW1ED3	0003	.201 (5.11)	.231 (5.87)	.130 (3.30)	.170 (4.32)	---	---	.310 (7.87)	---	.490 (12.45)	---	.173 (4.39)	.203 (5.16)	---	---	.164-32 UNC-2A	1.42
FL42GW1ED3	0004	.201 (5.11)	.231 (5.87)	.078 (1.98)	.108 (2.74)	.050 REF. (1.27)		.281 (7.14)	.343 (8.71)	.859 (21.82)	.921 (23.39)	.172 (4.37)	.202 (5.13)	---	---	.164-32 UNC-2A	2.2

1/ The letter "W" indicates the insertion loss versus frequency characteristics specified in table III.
2/ The letter "D" indicates the operating temperature range -55°C to +125°C.

Hardware dimensions.

Dash no.	Hex nut					Lockwasher					
	A		B		C	D		E		F	
	Min	Max	Min	Max	Thread	Min	Max	Min	Max	Min	Max
0001 thru 0004	.235 (5.97)	.265 (6.73)	.073 (1.85)	.078 (1.98)	.164-32 UNC-2B	.278 (7.06)	.288 (7.32)	.016 (.41)	.020 (.51)	.162 (4.11)	.172 (4.37)

FIGURE 1. Case and hardware dimensions and circuit configuration - Continued.

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Requirements:

Dimensions and configuration: See figure 1.

Weight: See table I.

Case: Metal.

Terminals: Solderable.

Operating temperature range:
-55°C to +125°C; temperature characteristic D.

Rated voltage: 100 V dc or 70 V rms, to 400 Hz over the rated temperature range of -55°C to +125°C for dash numbers 0001, 0002, and 0004 and 50 V dc for dash number 0003.

Rated current: 10 amperes, dc or ac (rms) for dash numbers 0001, 0002, and 0004,and 25 amperes, dc and IF current (no insertion loss change), 20 amperes(no overheating), 0.3 amperes RF current for dash number 0003.

Insertion loss characteristic: Symbol W (see table I).

TABLE I. Insertion loss versus frequency.

Dash number	Minimum no load insertion loss (db) in accordance with MIL-STD-220 at +25°C								
	1 MHz	5 MHz	10 MHz	20 MHz	50 MHz	100 MHz	200 MHz	500 MHz	1-10 GHz
0001	---	---	---	---	25	45	60	60	60
0002	---	---	---	---	25	45	60	60	60
0003	---	---	20	---	50	65	75	---	85
0004	10	---	28	---	---	41	44	---	60

(E) Seal: Not applicable.

Capacitance to ground: Method 305, MIL-STD-202.
Test frequency: 1,000 ±100 Hz.
Measured capacitance shall be at least 1,500 pF for dash numbers 0001 and 0002; 15,000 pF for dash number 0003; and 22,000 pF for dash number 0004.

Temperature rise: +25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202. The test voltage shall be 600 V dc for dash numbers 0001 and 0002, 150 V dc for dash number 0003 and 200 V dc for dash number 0004 applied for 1 to 5 seconds.
Charging current shall not exceed 50 mA.

Barometric pressure (reduced): Method 105, MIL-STD-202, test condition D for dash numbers 0001 and 0002, test condition B for dash numbers 0003 and 0004.
Dielectric withstanding voltage: In accordance with initial requirements, except that the test voltage shall be 125 percent of the rated dc voltage.

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Insulation resistance: Method 302, test condition A, MIL-STD-202.
Insulation resistance measured at +25°C between either terminal and the case shall be at least 10 gigohms for dash numbers 0001 and 0002, 5000 megohms for dash number 0003 and 10,000 megohms for dash number 0004.

Voltage drop: Not applicable.

Insertion loss: In accordance with MIL-F-15733.

Overload: In accordance with MIL-F-15733.
Measurements at +25°C after test:
Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: Method 211, test condition A, MIL-STD-202.
Applied force: 5 pounds for dash numbers 0001 and 0002, 2 pounds for dash numbers 0003 and 0004.

Flashpoint of impregnant or potting compound: Not applicable.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Thermal shock and immersion:
Thermal shock: Method 107, test condition A, MIL-STD-202, except that the temperature of step 3 shall be +125°C.
Immersion: Method 104, test condition A, MIL-STD-202.
Measurements after final cycle:
Dielectric withstanding voltage with 90 percent of initial voltage applied for 5 ±1 seconds, insulation resistance shall be not less than 1 gigohm for dash numbers 0001, 0002, and 0004, and not less than 30 percent of initial requirement for dash number 0003.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.
Electrical load: 100 percent of rated voltage between terminals and case. After completion of the shock test, filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Vibration, high frequency: Method 204, test condition D, MIL-STD-202.
Electrical load: Rated voltage, and rated current shall be applied continuously during test.
Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

Moisture resistance: Method 107, test condition A, (except temperature of step 3 shall be +125°C), MIL-STD-202, except that no measurement shall be made before and after cycling.
Filters shall then be tested in accordance with method 106, MIL-STD-202.
Initial measurements: Not applicable.
Polarization voltage: 100 V dc.
Measurements after 24-hour drying period:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.
Insulation resistance shall be not less than 1 gigohm for dash numbers 0001, 0002, and 0004, and not less than 30 percent of initial requirements for dash number 0003.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

Life: Method 108, test condition D, MIL-STD-202. Test temperature +125°C.

Test voltage: 1.2 times rated rms voltage for dash numbers 0001, 0002, and 0004, and 1.4 times rated dc voltage for dash number 0003.
Test current: 10 amperes for dash numbers 0001, 0002, and 0004 and 25 amperes for dash number 0003.

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Measurements after test:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.
Insulation resistance shall be not less than 1 gigohm for dash numbers 0001, 0002, and 0004 and not less than 30 percent of initial requirements for dash number 0003.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

Part or Identifying Number (PIN): M15733/28-(dash number from table I).

Marking: Filters shall be marked, as a minimum, with an abbreviated PIN, as shown on figure 2. Full marking, in accordance with MIL-F-15733, shall be marked on the unit package.

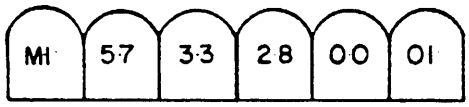


FIGURE 2. Example of marking for the abbreviated PIN on the hex flats - expanded view.

Supersession data: See table II.

Ⓔ TABLE II. Supersession data.

Superseded Part or Identifying Number (PIN)	Superseding Part or Identifying Number (PIN)
M15733/28-0001	M15733/61-0008
M15733/28-0002	M15733/61-0008
M15733/28-0003	M15733/61-0014
M15733/28-0004	M15733/61-0013

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activitits:
Army - AR, MI
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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